

Data Analysis and Machine Learning

Approaches for Predicting Movie Ratings

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1. Motivation

Movie ratings are important indicators of audience satisfaction and movie success. Platforms such as Netflix and IMDb contain large amounts of information about movies, including genres, release years, durations, audience votes, and ratings. Analyzing these variables can help identify which factors influence movie ratings the most.

The motivation of this project is to explore relationships between movie characteristics and IMDb scores using data analysis and machine learning methods. In addition to exploratory data analysis and hypothesis testing, several machine learning models were applied and compared to predict movie ratings based on movie features.

This project aims to understand audience behavior, discover meaningful patterns in movie datasets, and evaluate the performance of different machine learning algorithms.

2. Data Source

The project uses two public datasets:

- A Netflix movie dataset containing movie titles, genres, release years, and durations
- An IMDb dataset containing IMDb scores and audience vote counts

The datasets were merged using movie titles after preprocessing steps such as lowercase conversion and cleaning.

Main variables used in the project include:

- imdb_score
- num_voted_users
- release_year
- duration_min
- primary_genre

After merging and cleaning, missing values were removed and the dataset was prepared for analysis and machine learning applications.

3. Data Preparation

Several preprocessing steps were applied before analysis:

- Only movie entries were selected
- Titles were converted to lowercase
- Missing values were removed
- Movie durations were converted into numeric format
- Datasets were merged using movie titles
- The first listed genre was selected as the primary genre
- Categorical variables were transformed using one-hot encoding for machine learning models

These preprocessing steps improved dataset consistency and enabled effective analysis.

4. Exploratory Data Analysis

Exploratory Data Analysis was used to understand the structure of the dataset and identify relationships between variables.

First, the distribution of IMDb scores was analyzed. Most movies had medium to high IMDb scores, while very low scores were less common. This shows that the dataset mainly contains movies with moderate audience satisfaction.

Genre-based analysis showed that IMDb scores differ across movie genres. Some genres had higher average ratings than others, suggesting that genre may influence audience ratings.

The relationship between number of votes and IMDb score was also examined. The results showed a positive relationship, meaning that movies with more audience votes generally tended to have higher ratings.

Release year and duration were also analyzed. Older movies showed slightly higher average ratings in this dataset, while movie duration had a weaker relationship with IMDb score.

A correlation heatmap was also used to examine relationships between numerical variables. The heatmap showed that vote count had the strongest relationship with IMDb score compared to the other numerical features.

5. Hypothesis Testing

Three hypothesis tests were conducted to support the exploratory findings.

Hypothesis 1: Vote Count and IMDb Score

The first hypothesis tested whether there is a relationship between number of voted users and IMDb score.

- H0: There is no relationship between vote count and IMDb score.
- H1: There is a relationship between vote count and IMDb score.

The result showed a statistically significant positive relationship between vote count and IMDb score. This means that movies with more votes tend to have higher ratings.

Hypothesis 2: Genre and IMDb Score

The second hypothesis tested whether average IMDb scores differ across genres.

- H0: All genres have the same average IMDb score.
- H1: At least one genre has a different average IMDb score.

The result showed that genre has a statistically significant effect on IMDb score. This supports the idea that movie type can influence audience ratings.

Hypothesis 3: Old Movies vs New Movies

The third hypothesis tested whether older and newer movies have different average IMDb scores.

- H0: Older and newer movies have the same average IMDb score.
- H1: Older and newer movies have different average IMDb scores.

The result showed that older movies had slightly higher ratings in this dataset.

6. Machine Learning Analysis

Several machine learning models were applied to predict IMDb scores using movie-related features.

The target variable was:

- `imdb_score`

The feature variables included:

- `num_voted_users`
- `release_year`
- `duration_min`
- `primary_genre`

Since `primary_genre` is a categorical variable, one-hot encoding was applied before training the models.

The following models were used:

- Linear Regression
- Ridge Regression
- Random Forest Regressor
- Gradient Boosting Regressor

The models were evaluated using three metrics:

- RMSE
- MAE
- R^2 Score

RMSE and MAE measure prediction error. Lower values mean better performance. R^2 measures how much of the variation in IMDb scores is explained by the model. Higher R^2 values indicate better model performance.

7. Machine Learning Results

The model comparison showed that Random Forest performed better than the other models.

The results were:

Model	RMSE	MAE	R ²
Linear Regression	0.7887	0.6105	0.3048
Ridge Regression	0.7886	0.6103	0.3050
Random Forest	0.7553	0.5706	0.3624
Gradient Boosting	0.7797	0.5756	0.3205

Random Forest had the lowest RMSE and MAE, and the highest R² score. This suggests that nonlinear models are better at capturing relationships between movie features and IMDb ratings than simple linear models.

However, the R² values are still moderate. This means that the available features explain only part of the variation in IMDb scores. Movie ratings are also affected by many other factors that are not included in the dataset, such as actors, directors, marketing, cultural context, and personal preferences.

8. Feature Importance

Feature importance analysis was conducted using the Random Forest model.

The most important features were:

- num_voted_users
- duration_min
- release_year
- primary_genre_Dramas

The most important variable was the number of voted users. This shows that audience engagement is strongly related to IMDb ratings. Duration and release year also had some influence. Genre-related variables had smaller but still meaningful effects.

This result supports the earlier EDA and hypothesis testing findings, especially the relationship between vote count and IMDb score.

9. Findings

The main findings of this project are:

- Movies with more votes tend to have higher IMDb scores.
- Genre has a significant effect on movie ratings.
- Older movies have slightly higher ratings in this dataset.
- Duration has a weaker but visible relationship with IMDb score.
- Random Forest performed better than linear models for rating prediction.

- Vote count was the most important feature in the machine learning analysis.

Overall, both audience-related variables and movie characteristics influence IMDb ratings.

10. Limitations and Future Work

This project has some limitations.

First, the merged dataset became smaller after matching Netflix and IMDb movie titles. Some movies could not be matched because of differences in title formatting or missing values.

Second, the dataset includes limited features. Movie ratings may also depend on actors, directors, budget, language, country, marketing, and cultural factors. These variables were not included in this project.

Third, IMDb ratings are based on user votes, so they may contain audience bias. Movies with larger fan bases or stronger online visibility may receive more votes and higher ratings.

For future work, the dataset could be improved by adding more features such as director information, cast popularity, movie budget, country, language, and box office revenue. More advanced machine learning models could also be tested. Cross-validation and hyperparameter tuning could further improve model evaluation.

11. Conclusion

This project analyzed movie ratings using Netflix and IMDb datasets. The analysis included data collection, cleaning, exploratory data analysis, hypothesis testing, and machine learning.

The results showed that vote count, genre, release year, and duration are related to IMDb scores. Among the machine learning models, Random Forest achieved the best performance. Feature importance analysis showed that number of voted users was the strongest predictor of IMDb score.

In conclusion, movie ratings are influenced by both audience engagement and movie characteristics. Although the models provided useful insights, more detailed data would be needed to make stronger predictions.